



SIMT Engineering College
Dhaka University Affiliated Technology Unit
CSE 1201: Fundamentals of Programming
Incouse 1

Time: 2 hour

Marks: 25

(Write Any 4 Marks: 4*4= 16)

1. Explain The Basic Operations of Computer Components.
2. Write the Difference Between Compiler and Interpreter.
3. Explain the Compilation Process in C with Diagram.
4. Write Short Definition: Variable, Token, Literal. Explain the Types of Constants. What is the Format Specifier of Fractional Equivalent?
5. Write about Data Type Modifiers.
6. What #define and #include directive explains about?

***Must Answer:

9

What is the output of the Code?

```
#include <stdio.h>

int main() {
    int a = 10, b = 5;
    1 printf("%d\n", a * b);
    2 printf("%d\n", a / b);
    3 printf("%d\n", a % b);
    4 printf(" %d\n", a >= b);
    5 printf("%d\n", a <= b);
    6 printf(" %d\n", (a > b) && (b > 0));
    7 printf("%d\n", (a > b) || (b < 0));
    8 printf("%d\n", !(a > b));
    c += b;
    9 printf("%d\n", c);
    10 printf("%d\n", a++);
    11 printf("%d\n", a);
    12 printf("%d\n", ++a);
    13 int max = (a > b) ? a : b;
    14 printf(" %d\n\n", max);
    15 printf("a & b = %d\n", a & b);
    16 printf("a | b = %d\n", a | b);
    17 printf("a ^ b = %d\n", a ^ b);
    18 printf("~a = %d\n", ~a);
    19 printf("a << 1 = %d\n", a << 1);
    20 printf("a >> 1 = %d\n\n", a >> 1);
    21 printf("sizeof(a) = %zu bytes\n", sizeof(a));
    return 0;
}
```

SAIC Institutes of Management and Technology

Affiliated Engineering Colleges, University of Dhaka

Department of Computer Science & Engineering

1st Year 2nd Semester In course -1 Examination, 2025

Course Code: CSE 1202

Time: 50 min

Course Title: Digital Logic Design

[Answer All the Questions]

Marks: 5*5=25

- ✓ 1. Convert the binary code **101101** to Gray and vice-versa. 3
- ✓ 2. Implement a two input XOR gate using only two input NAND gates. 4
- ✓ 3. Find the 10's complement of **(325.625)₁₀** 3
- ✓ 4. Develop a truth table for the following SOP (Sum of Product) expression: $F(A, B, C) = AB + A'BC + A'C + ABC$ 5
- ✓ 5. Convert the expression into both SOP and POS canonical forms: $F(A, B, C) = (A+B)(A'+C)$ 5
- ✓ 6. Simplify the Boolean function using K-map $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$ 5

SIMT Engineering College

Department of CSE

In course Exam-1

Time: 60 minutes

Physics

Total-20

Answer all the following questions:

1. Describe interference and show that thickness of light and bright fringe are same in interference pattern. (2+8)
2. Explain^{why} the center of Newtons ring is dark? (5)
3. Write short note on:
(i) Huygens principle, (ii) Superposition theorem (2.5+2.5)

$$\sin(\alpha + \beta) \\ \sin \alpha \cos \beta + \cos \alpha \sin \beta$$



SIMT Engineering College
Dhaka University Affiliated Technology Unit
CSE 1204: Methods of Integration, Differential Equations and Series
Incourse 1

Time: 2 hour

Marks: 30

1. Evaluate the Integral $\int \frac{x + \sin x}{1 + \cos x} dx$.

2. show that

$$\int \frac{2 \sin x + 3 \cos x}{7 \sin x - 2 \cos x} dx = \frac{8x}{53} + \frac{25}{53} \ln (7 \sin x - 2 \cos x) + c.$$

3. Evaluate the Integral $\int \frac{dx}{3 + 2 \cos x}$

4. Find the area of a loop of $r = a \cos 3\theta$.

5. Find the area of a loop of the curve
 $r = a(1 + \cos \theta)$.

6. Find the area of a loop of the curve $r^2 = a^2 \cos 2\theta$.

Signature
03.09.2025
Lecturer: Mathematics
SIMT Engineering College

SAIC Institute of Management and Technology
Affiliated Engineering College
University of Dhaka
Department of CSE
Course name: Developing English Language Skills
Course code: ENG 1215
Marks: 40
Time: 1 Hour 20 minutes

Read the passage and answer question no 1 & 2.

Sea monsters are the stuff of legend - lurking not just in the depths of the oceans, but also the darker corners of our minds. What is it that draws us to these creatures?

"This inhuman place makes human monsters," wrote Stephen King in his novel *The Shining*. Many academics agree that monsters lurk in the deepest recesses, they prowl through our ancestral minds appearing in the half-light, under the bed - or at the bottom of the sea.

"They don't really exist, but they play a huge role in our mindscapes, in our dreams, stories, nightmares, myths and so on," says Matthias Classen, assistant professor of literature and media at Aarhus University in Denmark, who studies monsters in literature. "Monsters say something about human psychology, not the world."

One Norse legend talks of the Kraken, a deep-sea creature that was the curse of fishermen. If sailors found a place with many fish, most likely it was the monster that was driving them to the surface. If it saw the ship, it would pluck the hapless sailors from the boat and drag them to a watery grave.

This terrifying legend occupied the mind and pen of the poet Alfred Lord Tennyson too. In his short 1830 poem *The Kraken* he wrote: "Below the thunders of the upper deep, / Far far beneath in the abysmal sea, / His ancient, dreamless, uninvaded sleep / The Kraken sleepeth."

The deeper we travel into the ocean, the deeper we delve into our own psyche. And when we can go no further - there lurks the Kraken.

Most likely the Kraken is based on a real creature - the giant squid. The huge mollusc takes pride of place as the personification of the terrors of the deep sea. Sailors would have encountered it at the surface, dying, and probably thrashing about. It would have made a weird sight, "about the most alien thing you can imagine," says Edith Widder, CEO at the Ocean Research and Conservation Association.

"It has eight lashing arms and two slashing tentacles growing straight out of its head and it's got serrated suckers that can latch on to the slimiest of prey and it's got a parrot beak that can rip flesh. It's got an eye the size of your head, it's got a jet propulsion system and three hearts that pump blue blood."

The giant squid continued to dominate stories of sea monsters with the famous 1870 novel, *Twenty Thousand Leagues Under the Sea*, by Jules Verne. Verne's submarine fantasy is a classic story of puny man against a gigantic squid.

The monster needed no embellishment - this creature was scary enough, and Verne incorporated as much fact as possible into the story, says Emily Alder from Edinburgh Napier University. "*Twenty Thousand Leagues Under the Sea* and another contemporaneous book, Victor Hugo's *Toilers of the Sea*, both tried to represent the giant squid as they might have been actual zoological animals, much more

taking the squid as a biological creature than a mythical creature." It was a given that the squid was vicious and would readily attack humans given the chance.

1. Answer the following questions

(1×5=5)

- a. Why do sea monsters like the Kraken continue to fascinate human imagination?
- b. How does the Norse legend describe the Kraken's interaction with sailors?
- c. What did Alfred Lord Tennyson describe in his poem *The Kraken*?
- d. Why did sailors find the giant squid terrifying?
- e. How did Jules Verne and Victor Hugo portray the giant squid in their novels?

2. Write a summary of the given text.

5

Grammar

A) Complete the following exercise with a / an or the. Leave blank if no article is needed.

10

i. Mecca is an ancient (a) — city. (b) — first mosque of (c) — world was built there. Hazrat Ibrahim built it. It is known as (d) — Kaaba Mosque. Many days have passed away. Suddenly a hole was found in (e) — wall. Today people all over (f) — world go to Mecca, they spend a busy (g) — time there. They forget the activities of this (h) — world. The Muslim of (i) — whole world gathers there and perform (j) — Hajj.

ii. English is (a) — international language. We feel (b) — necessity of learning English at every step. This is (c) — only language of international overseas business, communication, co-operation and (d) — co-existence. It is essential to receive (e) — foreign degree or higher education. All (f) — good jobs need English knowing persons. So, every educated man should know how to (g) — communicate in English. To learn and develop (h) — skill of English especially for (i) — students is very important. No (j) — student should be ignorant of it.

B) Write an email to the principal of your school requesting permission to arrange a cultural program.

10

C) Write a descriptive essay on "My favorite childhood memory"

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